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<https://git.aspp.school/ASPP/2026-prague-scientific-patterns>

Scientific programming patterns

Lisa Schwetlick, Pietro Berkes

What is wrong with you?!

- You studied the language
- You learned the libraries
- You coded for months
- And yet the code still feels like a 7-headed apocalyptic monster. Changes are painful, new features break old functionality, reproducing previous results becomes a git-checkout juggling exercise



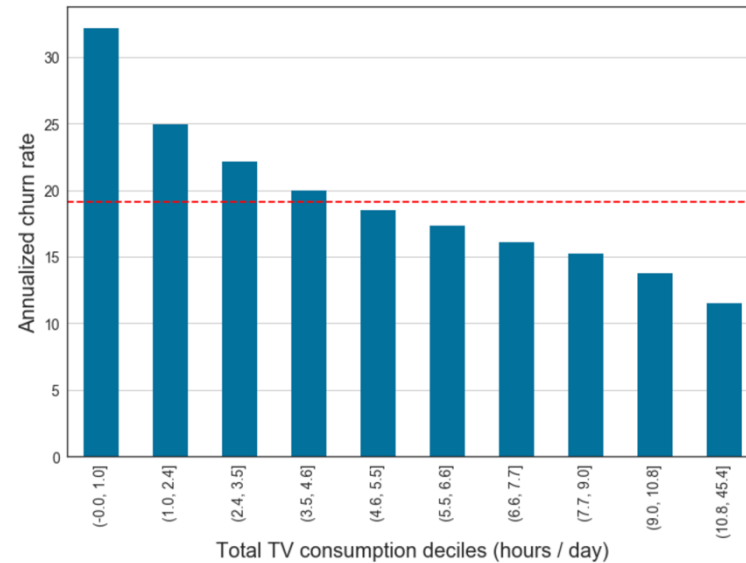
The good news: you can smell it

7.1 Total TV

```
In [118]: ttv_h_deciles = pd.qcut(ttv_h, 10)
ttv_deciles_churn = tv_merged.groupby(ttv_h_deciles).churned_all.mean() * 12 / 10 * 100
```

```
In [119]: with plt.rc_context(rc=get_style(figsize=(12, 8))):
    ax = ttv_deciles_churn.plot.bar(color=blue)
    plt.grid(axis='y')
    plt.xlabel('Total TV consumption deciles (hours / day)')
    plt.ylabel('Annualized churn rate')
    plt.axhline(annual_tv_churn * 100, c='r', ls='--')

    t = ['({:.1f}, {:.1f})'.format(x.left, x.right) for x in ttv_deciles_churn.index]
    plt.xticks(range(len(t)), t)
```



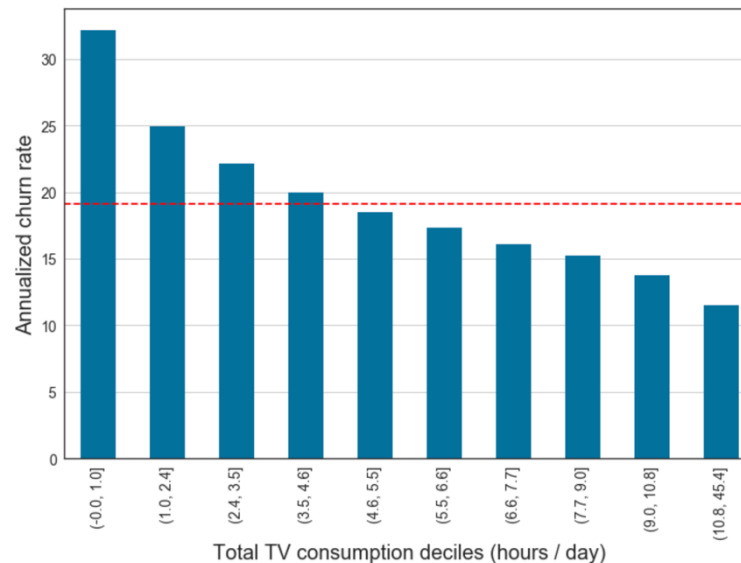
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7.1 Total TV

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```
In [119]: with plt.rc_context(rc=get_style(figsize=(12,8))):
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    plt.ylabel('Annualized churn rate')
    plt.axhline(annual_tv_churn * 100, c='r', ls='--')

    t = ['({:.1f}, {:.1f})'.format(x.left, x.right) for x in ttv_deciles_churn.index]
    plt.xticks(range(len(t)), t)
```



7.2 Playback

```
In [120]: replay_deciles = pd.qcut(tv_merged.PLAYBACK / ttv_sec_to_hpd, 10)
replay_deciles_churn = tv_merged.groupby(replay_deciles).churned_all.mean() * 12 / 10 * 100
```

```
In [121]: with plt.rc_context(rc=get_style(figsize=(12,8))):
    ax = replay_deciles_churn.plot.bar(color=blue)
    plt.grid(axis='y')
    plt.xlabel('Replay TV consumption deciles (hours / day)')
    plt.ylabel('Annualized churn rate')
    plt.axhline(annual_tv_churn * 100, c='r', ls='--')

    t = ['({:.1f}, {:.1f})'.format(x.left, x.right) for x in replay_deciles_churn.index]
    plt.xticks(range(len(t)), t)
```

```
In [122]: replay_h = pd.cut(tv_merged.PLAYBACK / ttv_sec_to_hpd, np.arange(0, 8), include_lowest=True)
replay_churn = tv_merged.groupby(replay_h).churned_all.mean() * 12 / 10 * 100
```

```
In [123]: with plt.rc_context(rc=get_style(figsize=(12,8))):
    ax = replay_churn.plot.bar(color=blue)
    plt.grid(axis='y')
    plt.xlabel('Replay TV consumption (hours / day)')
    plt.ylabel('Annualized churn rate')
    plt.axhline(annual_tv_churn * 100, c='r', ls='--')
```

7.3 Trends

```
In [137]: tv_delta = pd.cut(tv_merged.TTV_201703_delta, np.arange(-5.5, 5.6, 1.0))
tv_delta_churn = tv_merged.groupby(tv_delta).churned_all.mean() * 12 / 10 * 100
```

```
In [138]: with plt.rc_context(rc=get_style(figsize=(12,8))):
    ax = tv_delta_churn.plot.bar(color=blue)
    plt.grid(axis='y')
    plt.xlabel('TV consumption trend')
    plt.ylabel('Annualized churn rate')
    plt.axhline(annual_tv_churn * 100, c='r', ls='--')
```

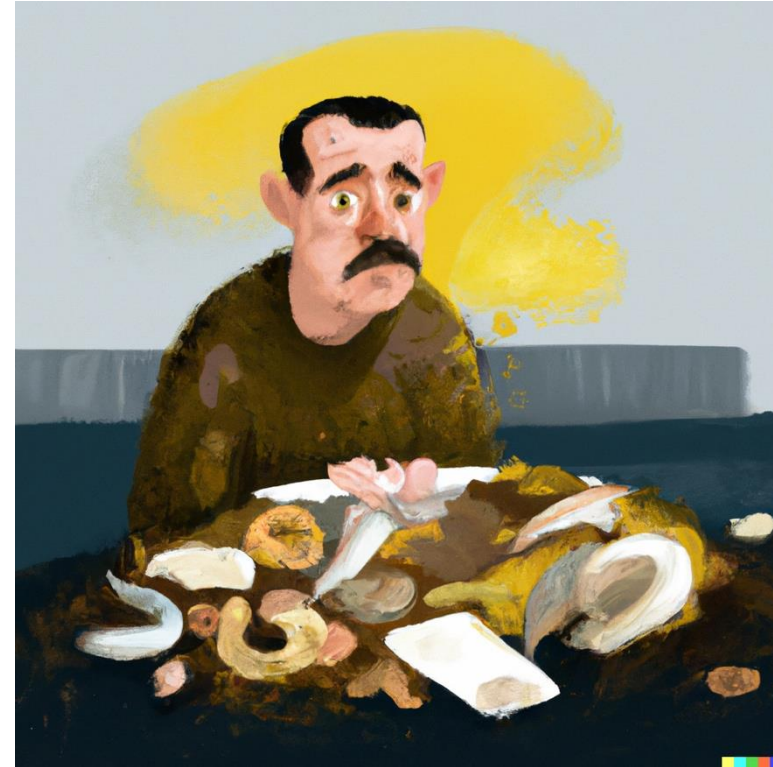
```
In [1651]: tv_delta = pd.cut(tv_merged.TTV_201703_delta, np.arange(-5.5, 5.6, 1.0))
tv_delta_churn_some = tv_merged[at_least_some_ttv_mask].groupby(tv_delta[at_least_some_ttv_mas
```

```
In [1652]: with plt.rc_context(rc=get_style(figsize=(12,8))):
    ax = tv_delta_churn_some.plot.bar(color=blue)
    plt.grid(axis='y')
    plt.xlabel('TV consumption trend')
    plt.ylabel('Annualized TV+Vodafone churn rate')
    plt.axhline(annual_tv_churn * 100, c='r', ls='--')
```

What is wrong with smelly code?

Smelly code might work right now, but in time it is going to have one or more of these issues:

- **Hard to read and test:** it is difficult to see an overall structure; understanding the code in one place requires checking other code all over
- **Coupled:** an update in one place requires several other changes in other places
- **Not flexible:** adding new functionality and modifying exiting features require extensive rewrites or hacks



Code should be as a construction block structure

Flexible code is just like a block construction:

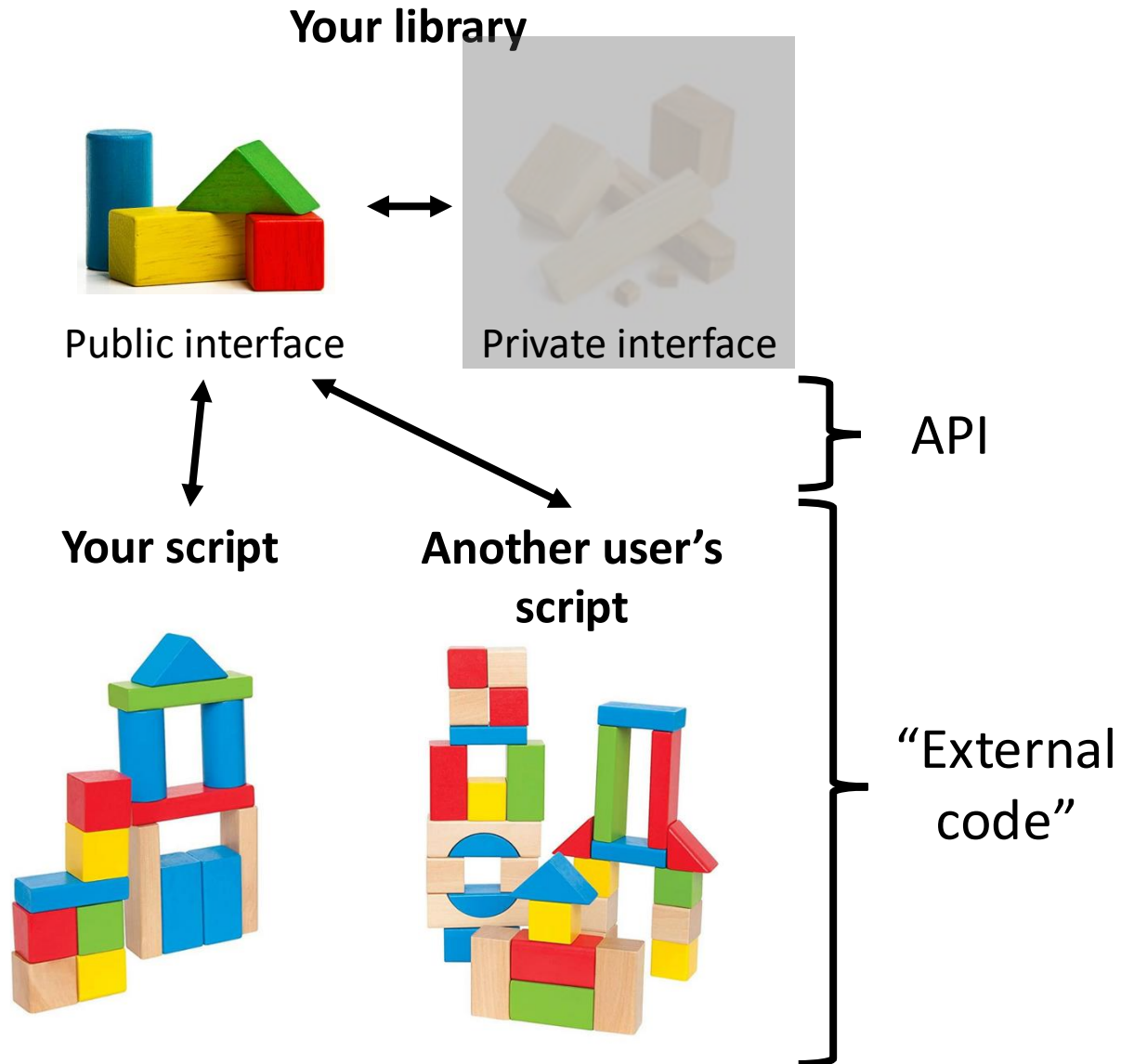
- is easy to understand in terms of blocks
- tolerates changes
- is reusable

Functions and classes group together things that are coupled together and form the basic blocks.

We can easily and quickly rearrange the blocks to extend a structure or build a new one!



Words that we'll say a lot



- Library vs external code
 - External code is anything that is not in your library: your script, 3rd party wanting to do other stuff, another library
- Interface, API
 - How you are supposed to call your library. The list of public functions and classes, and how to use them
- Public interface vs private interface
 - Changing the public interface forces all the external code to change

Chapter 1: Introduction to classes

Put together things that belong together



Classes live coding



Live Coding

notebooks/01a_Classes.ipynb

Hands-on



Exercise

exercises/particle_update_position

Add a new method to the Particle class

- Make the function `update_position` into a method of the class `Particle`.
- Where does the `position` of the particle belong? Modify the class constructor if necessary.
- Submit a PR for Issue #1 on [git.aspp.school](https://github.com/aspp/school)

Summary: Classes organize your code

```
def first_function(x, y, z):  
    # Something
```

```
def second_function(x, y, z):  
    # Something else
```

```
def third_function(x, y, z):  
    # Something more
```

... becomes ...

```
class Xyz:  
    def __init__(self, x, y, z):  
        ...  
    def first_function(self):  
        # Something  
    def second_function(self):  
        # Something else  
    def third_function(self):  
        # Something more
```

- Classes help us group data and functionalities that belong together
- When used judiciously, the code becomes more usable as we get rid of a lot of manual book-keeping; details are hidden away
- Understanding what belongs to the class and what does not is important to keep the code flexible!

Recap: Class structure

```
class MyClass:
```

```
    def __init__(self, param1, param2):  
        self.param1 = param1  
        if param2 is None:  
            param2 = 12.3  
        self.param2 = param2
```

The constructor is used to first populate an instance, called by convention “self”

```
    def my_method(self, foo, bar):  
        result = self.param2 * foo + self.param2 * bar  
        return result
```

```
    def to_json(self):  
        params = {  
            'param1': self.param1,  
            'param2': self.param2,  
        }  
        return json.dumps(params)
```

Classes can define “methods”, i.e. functions that have access to the data stored in an instance

```
@classmethod
```

```
def from_json(cls, json_str):  
    json_dict = json.loads(json_str)  
    instance = cls(json_dict['params1'], json_dict['params2'])  
    return instance
```

A “class method” can be used to build an instance in some alternative way, e.g. using data from a file

```
instance = MyClass(5, 18)  
instance = MyClass.from_json(json_str)
```

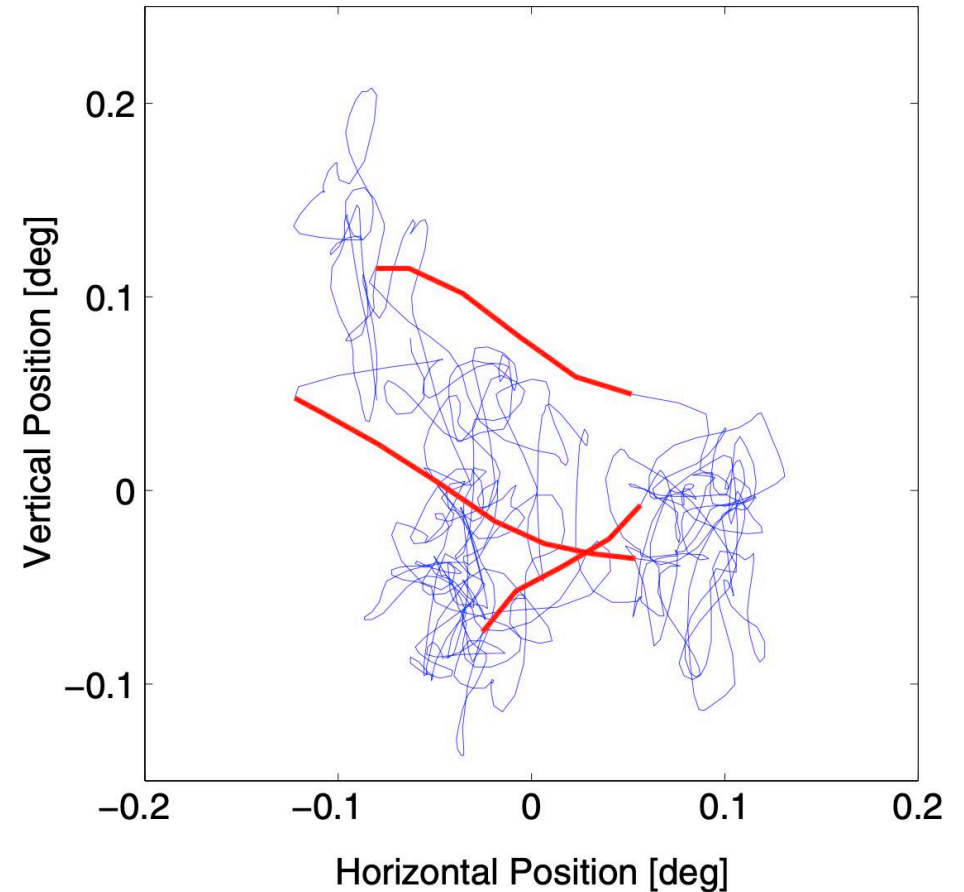
Here is how you create instances from the constructor or a class method

Chapter 2: Break out things that vary independently

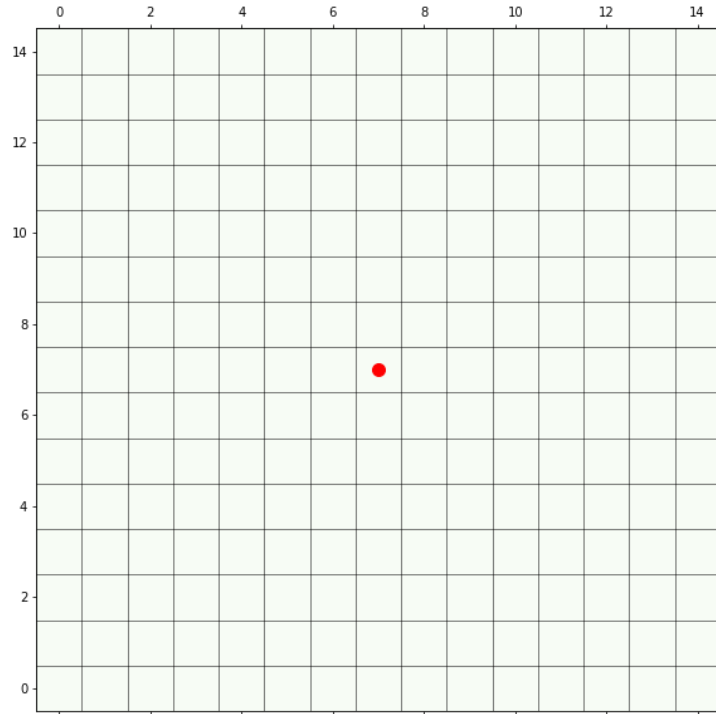
Excursion: let's walk!



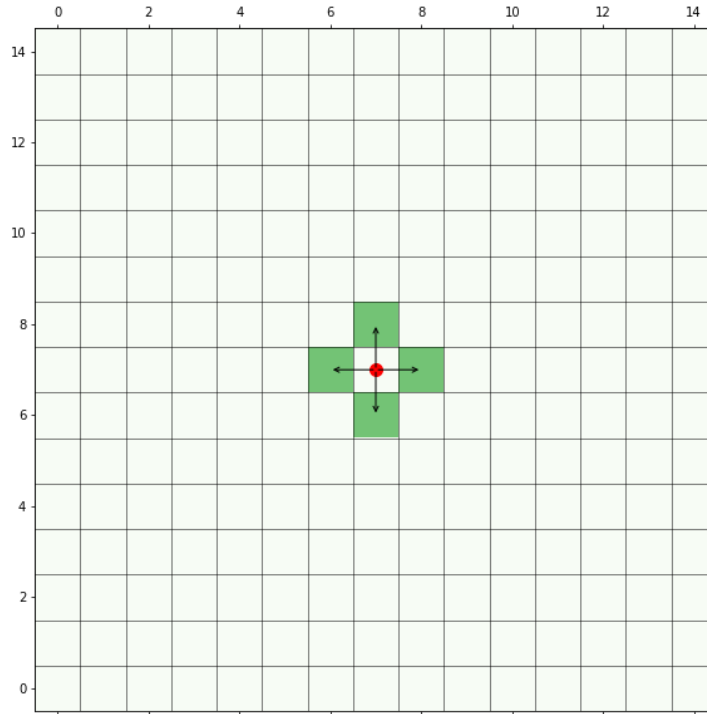
- Even when people think they are holding their eyes still, they still move around:
 - Drift, microsaccades, jitter
- The movement has statistical properties such as self avoidance, directional persistence...
- The upcoming exercise is a simplified version of a model Lisa worked on



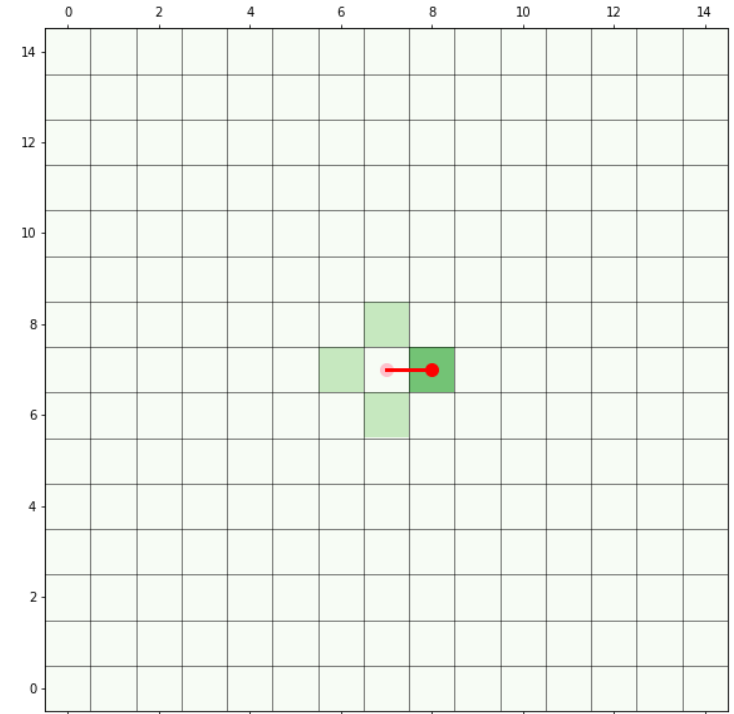
Excursion: let's walk!



Walker starts
somewhere

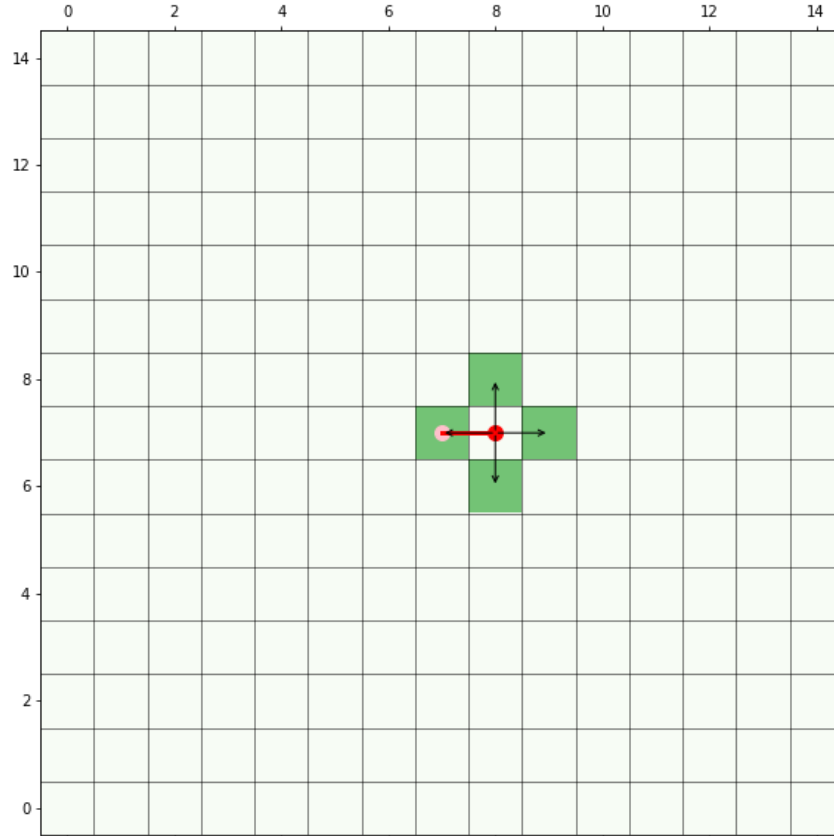


It could walk
one step in any
direction

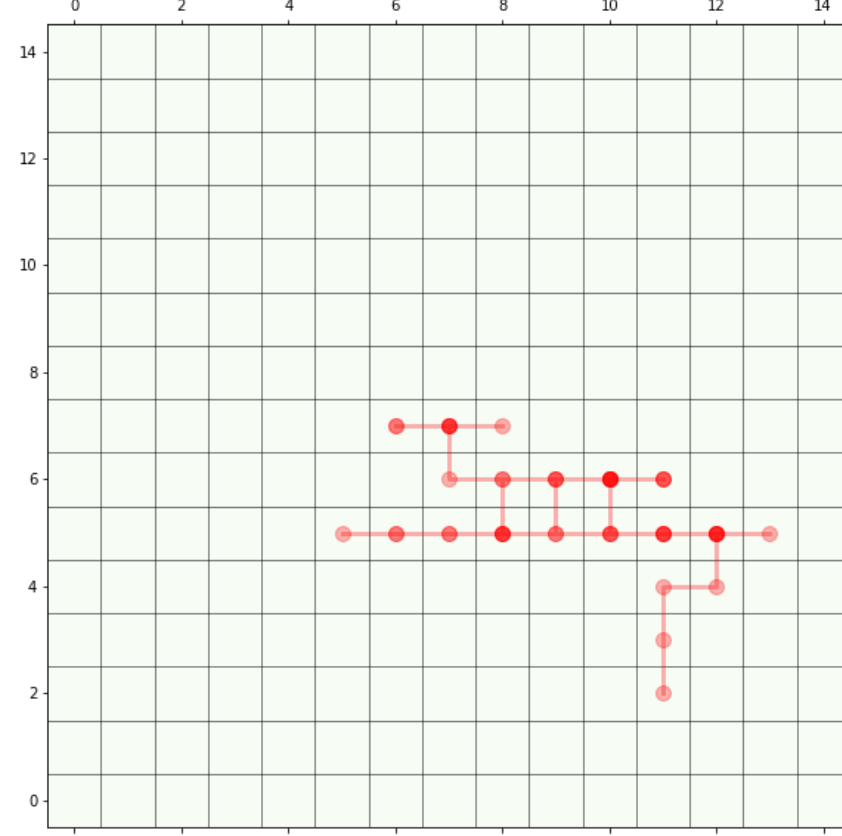


It randomly
selects one step

Excursion: let's walk!



In the next step it has the same stepping options

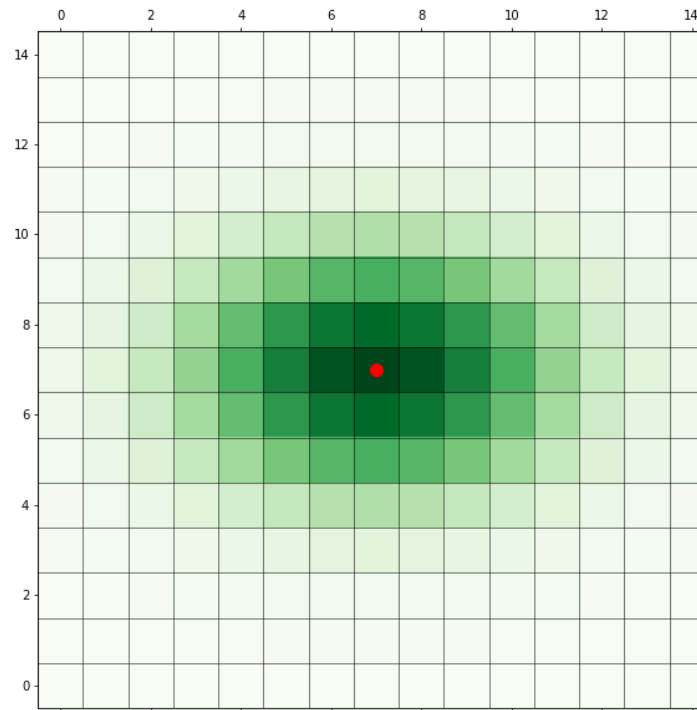


By iterating this procedure, we get a trajectory

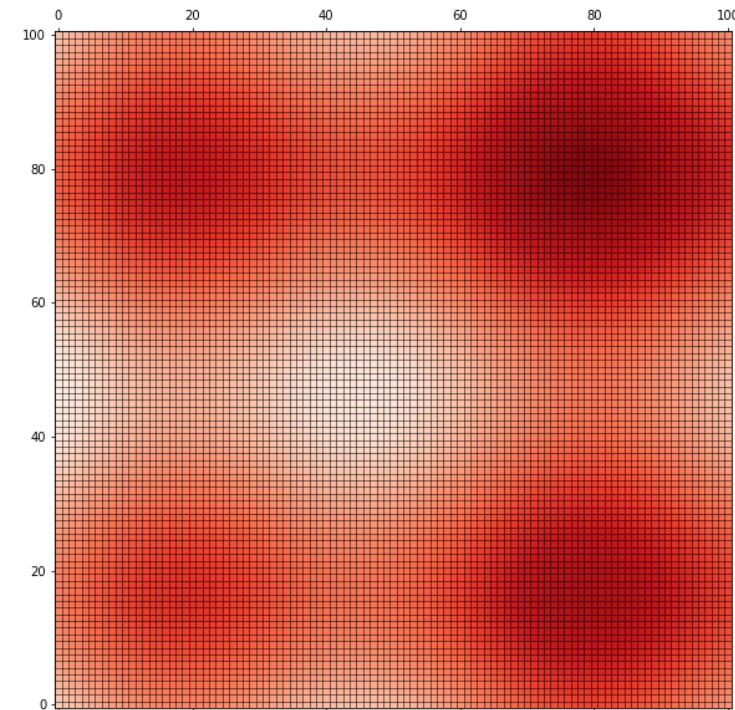
Excursion: let's walk!



- To make the behavior a little bit more interesting the walker in the exercise...
 - Choses its next step from a probability distribution
 - Can also walk over a non-uniform background, that influences how likely it is to go there



Stepping Probability



Background Activation

The walker Functions



Live Coding

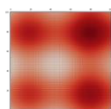
walker/Step_0_Introduction/

```
next_step_proposal(current_i, current_j, sigma_i, sigma_j, size)
```

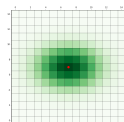
```
create_context_map(size, map_type='flat')
```

```
sample_next_step(current_i, current_j, sigma_i, sigma_j, context_map,  
random_state=np.random)
```

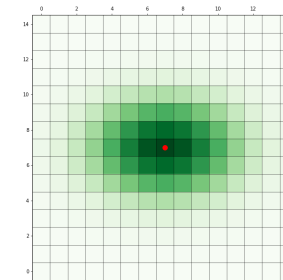
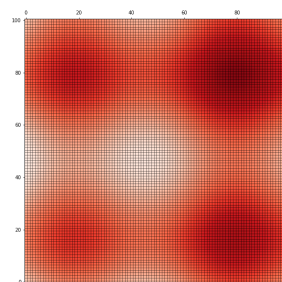
Combines



and



, and probabilistically selects the next location



Conceptual Exercises

- All the exercises are about **reformatting and pseudocode!** You are not expected to go into detail about how the code works- copy&pasting is fine!
- There are many ways to solve the exercises. **It is more important to think about the implementation choices with your partner than to finish.**
- We will give you working code to start from at each step, so don't worry about finishing!

Conceptual Exercises



Exercise

walker/Step_1_classes/

- We DON'T expect a PR, or working code! Instead, decide what would be a good way to build the code and **comment** your opinion in the git.aspp.school issue.

```
# Markdown Title
```

```
Comment in words, what you think!
```

```
```python
```

```
S = "The maintainers of the library are awesome!"
```

```
print(s)
```

```
```
```

- Open the notebook "Step_1_classes_exercise" execute the code, and follow the instructions
- Comment on Issue #2 on git.aspp.school, suggesting what you think the code should look like.

Plotting does not belong to the Walker



Live Coding

walker/Step_2_plotting/

```
import numpy as np
import matplotlib.pyplot as plt

class Walker:
    """ The Walker knows how to walk at random on a context map. """

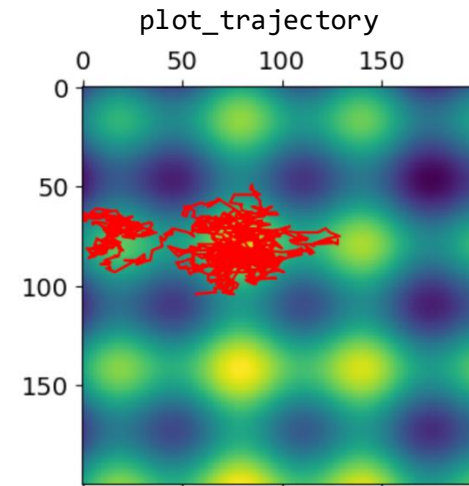
    def __init__(self, sigma_i, sigma_j, size, map_type='flat'):
        # ...

    def plot_trajectory(self, trajectory):
        """ Plot a trajectory over a context map. """
        trajectory = np.asarray(trajectory)
        plt.matshow(self.context_map)
        plt.plot(trajectory[:, 1], trajectory[:, 0], color='r')
        plt.show()

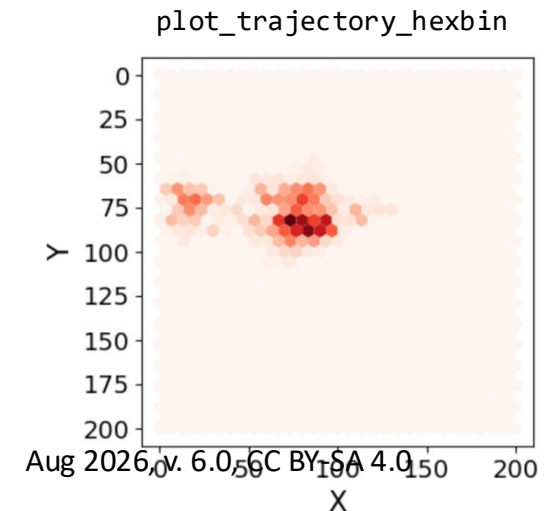
    def plot_trajectory_hexbin(self, trajectory):
        """ Plot an hexagonal density map of a trajectory. """
        trajectory = np.asarray(trajectory)
        plt.hexbin(
            trajectory[:, 1], trajectory[:, 0],
            gridsize=30, extent=(0, 200, 0, 200),
            edgecolors='none', cmap='Reds'
        )
        plt.gca().invert_yaxis()
        plt.xlabel('X')
        plt.ylabel('Y')
```

1. Changing or adding new way of plotting (e.g. by a colleague) would require modifying the Walker's code, without changing its behavior
2. Changing the Walker behavior will typically not modify the plotting code
3. Using your walker requires you to install matplotlib

These are smells of the fact that plotting varies independently of the Walker



vs.



The smells of the Walker constructor

```
def __init__(self, sigma_i, sigma_j, size, map_type='flat'):
    self.sigma_i = sigma_i
    self.sigma_j = sigma_j
    self.size = size

    if map_type == 'flat':
        context_map = np.ones((size, size))
    elif map_type == 'hills':
        grid_ii, grid_jj = np.mgrid[0:size, 0:size]
        i_waves = np.sin(grid_ii / 130) + np.sin(grid_ii / 10)
        i_waves /= i_waves.max()
        j_waves = np.sin(grid_jj / 100) + np.sin(grid_jj / 50) + \
            np.sin(grid_jj / 10)
        j_waves /= j_waves.max()
        context_map = j_waves + i_waves
    elif map_type == 'labyrinth':
        context_map = np.ones((size, size))
        context_map[50:100, 50:60] = 0
        context_map[20:89, 80:90] = 0
        context_map[90:120, 0:10] = 0
        context_map[120:size, 30:40] = 0
        context_map[180:190, 50:60] = 0

        context_map[50:60, 50:200] = 0
        context_map[179:189, 80:130] = 0
        context_map[110:120, 0:190] = 0
        context_map[120:size, 30:40] = 0
        context_map[180:190, 50:60] = 0
    context_map /= context_map.sum()
    self.context_map = context_map

    # Pre-compute a 2D grid of coordinates for efficiency
    self._grid_ii, self._grid_jj = np.mgrid[0:size, 0:size]
```

The smells of the Walker constructor

```
def __init__(self, sigma_i, sigma_j, size, map_type='flat'):  
    self.sigma_i = sigma_i  
    self.sigma_j = sigma_j  
    self.size = size  
  
    if map_type == 'flat':  
        context_map = np.ones((size, size))  
    elif map_type == 'hills':  
        grid_ii, grid_jj = np.mgrid[0:size, 0:size]  
        i_waves = np.sin(grid_ii / 130) + np.sin(grid_ii / 10)  
        i_waves /= i_waves.max()  
        j_waves = np.sin(grid_jj / 100) + np.sin(grid_jj / 50) + \n            np.sin(grid_jj / 10)  
        j_waves /= j_waves.max()  
        context_map = j_waves + i_waves  
    elif map_type == 'labyrinth':  
        context_map = np.ones((size, size))  
        context_map[50:100, 50:60] = 0  
        context_map[20:89, 80:90] = 0  
        context_map[90:120, 0:10] = 0  
        context_map[120:size, 30:40] = 0  
        context_map[180:190, 50:60] = 0  
  
        context_map[50:60, 50:200] = 0  
        context_map[179:189, 80:130] = 0  
        context_map[110:120, 0:190] = 0  
        context_map[120:size, 30:40] = 0  
        context_map[180:190, 50:60] = 0  
    context_map /= context_map.sum()  
    self.context_map = context_map  
  
    # Pre-compute a 2D grid of coordinates for efficiency  
    self._grid_ii, self._grid_jj = np.mgrid[0:size, 0:size]
```

1. The constructor will become longer with more map types
2. We cannot contribute a new map type without modifying the code of the Walker
3. It is difficult to test
4. It is not flexible, e.g. what happens if we want to create an instance from a context map saved on file?
5. An external user cannot add their own map type

These are smells of the fact that the initialization of context_map varies independently of the Walker

if ... elif ... else chain is often the smell of independent plug-in blocks

```
def add_block(blocks, size, type):  
    if type == 'circle':  
        # build circle of given size  
        block =   
    elif type == 'square':  
        # build square of given size  
        block =   
    elif type == 'triangle':  
        # build triangle of given size  
        block =   
    blocks.append(block)
```

```
add_block(blocks, size=11,  
          type='square')
```

if ... elif ... else chain is often the smell of independent plug-in blocks

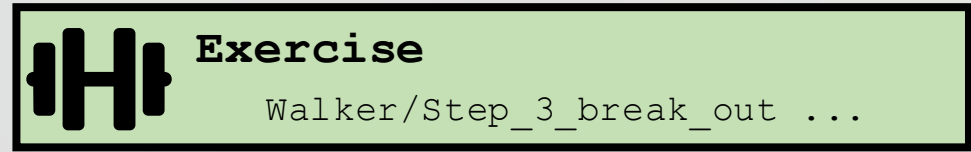
```
def add_block(blocks, size, type):  
    if type == 'circle':  
        # build circle of given size  
        block =   
    elif type == 'square':  
        # build square of given size  
        block =   
    elif type == 'triangle':  
        # build triangle of given size  
        block =   
    blocks.append(block)
```

```
add_block(blocks, size=11,  
          type='square')
```

```
def build_circle(size):  
    """ Build a circle. """  
    return   
  
def build_square(size):  
    """ Build a square. """  
    return   
  
def build_triangle(size):  
    """ Build a triangle. """  
    return   
  
def add_block(blocks, block):  
    blocks.append(block)
```

```
block = build_square(size=11)  
add_block(blocks, block)
```

Conceptual Exercise



Moving context map creation to separate modules

- Go to the Notebook and follow the instructions there
- Comment on Issue #3 on git.aspp.school, suggesting what you think the code should look like.

New requirement: we need to use different next-step proposals for different experiments

```
class Walker:
    # ...

    def sample_next_step(self, current_i, current_j, random_state=np.random):
        """ Sample a new position for the walker. """
        # Combine the next-step proposal with the context map to get a
        # next-step probability map
        next_step_map = self._next_step_proposal(current_i, current_j)
        selection_map = self._compute_next_step_probability(next_step_map)

        # Draw a new position from the next-step probability map
        r = random_state.rand()
        cumulative_map = np.cumsum(selection_map)
        cumulative_map = cumulative_map.reshape(selection_map.shape)
        i_next, j_next = np.argwhere(cumulative_map >= r)[0]

        return i_next, j_next

    def _next_step_proposal(self, current_i, current_j):
        """ Create the 2D proposal map for the next step of the walker. """
        # 2D Gaussian distribution, centered at current position,
        # and with different standard deviations for i and j
        grid_ii, grid_jj = self._grid_ii, self._grid_jj
        sigma_i, sigma_j = self.sigma_i, self.sigma_j

        rad = (
            (((grid_ii - current_i) ** 2) / (sigma_i ** 2))
            + (((grid_jj - current_j) ** 2) / (sigma_j ** 2))
        )

        p_next_step = np.exp(-(rad / 2.0)) / (2.0 * np.pi * sigma_i * sigma_j)
        return p_next_step / p_next_step.sum()
```



- We would like to run some experiments with a Gaussian next step proposal, some with a rectangular proposal, etc.
- **How can we do that?**

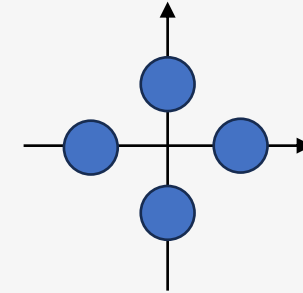
The inheritance solution

```
class Walker:
    # ...

    def _next_step_proposal(self, current_i, current_j):
        """ Create the 2D proposal map for the next step of the walker. """
        raise NotImplementedError("`_next_step_proposal` not implemented")
```

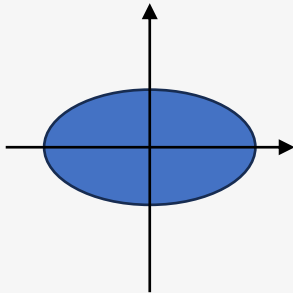
```
class JumpingWalker(Walker):
    # ...

    def _next_step_proposal(self, current_i, current_j):
```



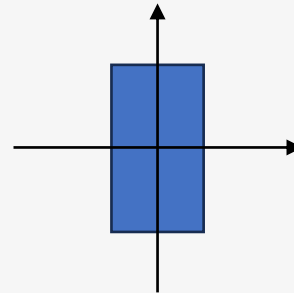
```
class GaussianWalker(Walker):
    # ...

    def _next_step_proposal(self, current_i, current_j):
```



```
class RectangularWalker(Walker):
    # ...

    def _next_step_proposal(self, current_i, current_j):
```



The problem with inheritance

New requirement: the way of combining next step proposal and context map can also vary

```
class Walker:
    # ...

    def _compute_next_step_probability(self, next_step_map):
        """ Compute the next step probability map from next step proposal and
        context map. """
        next_step_probability = next_step_map * self.context_map
        next_step_probability /= next_step_probability.sum()
        return next_step_probability
```

This combination can now be done through product, sum, or other

The problem with inheritance

The inheritance approach leads to a **combinatorial explosion of subclasses**

Gaussian
Rectangular $\times$ Product
Jumping Sum

```
class GaussianWalkerWithProductInteraction(Walker):
    def _next_step_proposal(self, current_i, current_j):
        # ...
    def _compute_next_step_probability(self, next_step_map):
        # ...

class GaussianWalkerWithSumInteraction(Walker):
    def _next_step_proposal(self, current_i, current_j):
        # ...
    def _compute_next_step_probability(self, next_step_map):
        # ...

class RectangularWalkerWithProductInteraction(Walker):
    def _next_step_proposal(self, current_i, current_j):
        # ...
    def _compute_next_step_probability(self, next_step_map):
        # ...

class RectangularWalkerWithSumInteraction(Walker):
    def _next_step_proposal(self, current_i, current_j):
        # ...
    def _compute_next_step_probability(self, next_step_map):
        # ...
```

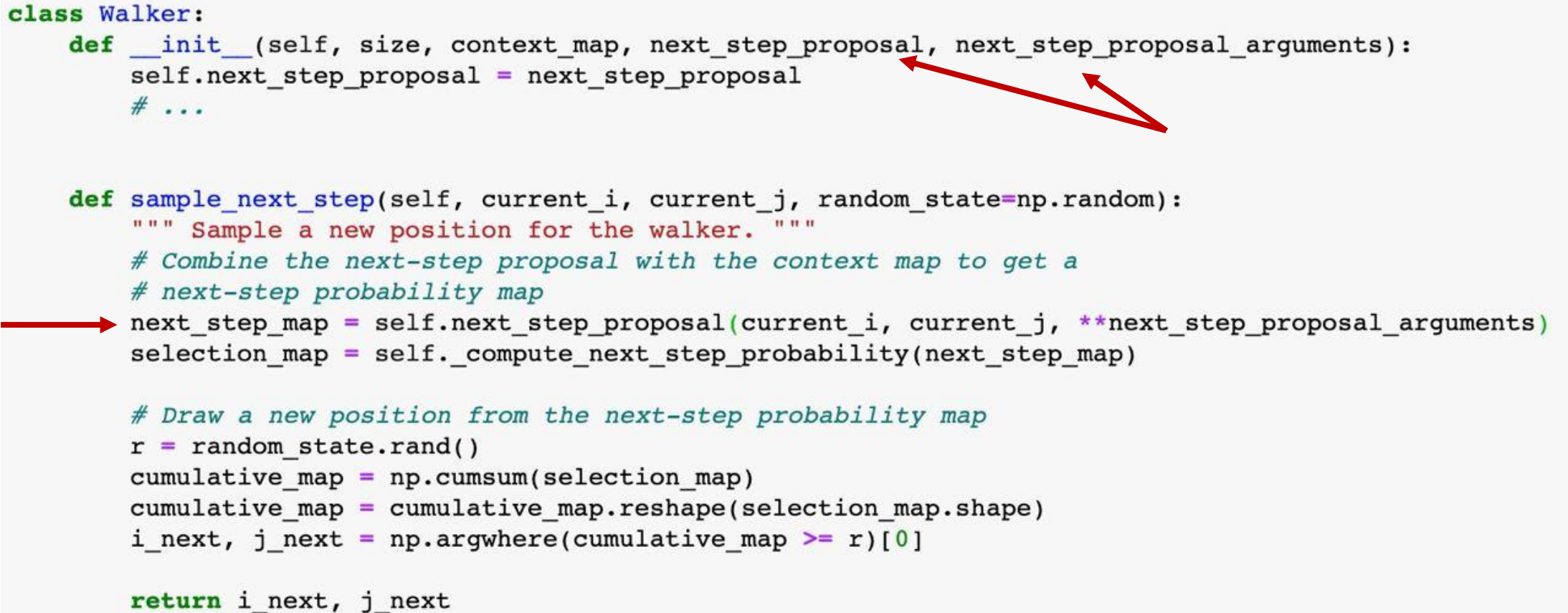
Passing varying behavior (e.g. functions) as an argument is usually a better alternative

```
class Walker:
    def __init__(self, size, context_map, next_step_proposal, next_step_proposal_arguments):
        self.next_step_proposal = next_step_proposal
        # ...

    def sample_next_step(self, current_i, current_j, random_state=np.random):
        """ Sample a new position for the walker. """
        # Combine the next-step proposal with the context map to get a
        # next-step probability map
        next_step_map = self.next_step_proposal(current_i, current_j, **next_step_proposal_arguments)
        selection_map = self._compute_next_step_probability(next_step_map)

        # Draw a new position from the next-step probability map
        r = random_state.rand()
        cumulative_map = np.cumsum(selection_map)
        cumulative_map = cumulative_map.reshape(selection_map.shape)
        i_next, j_next = np.argwhere(cumulative_map >= r)[0]

        return i_next, j_next
```



In this way, we can **define the behavior independently** of the class, avoiding the combinatoric explosion

if ... elif ... else chain is often the smell of independent plug-in blocks

```
def add_block(blocks, size, type):  
    if type == 'circle':  
        # build circle of given size  
        block =   
    elif type == 'square':  
        # build square of given size  
        block =   
    elif type == 'triangle':  
        # build triangle of given size  
        block =   
    blocks.append(block)
```

```
add_block(blocks, size=11,  
          type='triangle')
```

if ... elif ... else chain is often the smell of independent plug-in blocks

```
def add_block(blocks, size, type):  
    if type == 'circle':  
        # build circle of given size  
        block =   
    elif type == 'square':  
        # build square of given size  
        block =   
    elif type == 'triangle':  
        # build triangle of given size  
        block =   
    blocks.append(block)
```

```
add_block(blocks, size=11,  
          type='triangle')
```

```
def build_circle(size):  
    """ Build a circle. """  
    return   
  
def build_square(size):  
    """ Build a square. """  
    return   
  
def build_triangle(size):  
    """ Build a triangle. """  
    return   
  
def add_block(blocks, block):  
    blocks.append(block)
```

```
block = build_triangle(size=11)  
add_block(blocks, block)
```

if ... elif ... else chain is often the smell of independent plug-in blocks

```
def add_block(blocks, size, type):  
    if type == 'circle':  
        # build circle of given size  
        block =   
    elif type == 'square':  
        # build square of given size  
        block =   
    elif type == 'triangle':  
        # build triangle of given size  
        block =   
    blocks.append(block)
```

```
add_block(blocks, size=11,  
          type='square')
```

```
def build_circle(size):  
    """ Build a circle. """  
    return   
  
def build_square(size):  
    """ Build a square. """  
    return   
  
def build_triangle(size):  
    """ Build a triangle. """  
    return 
```

```
def add_block(blocks, size, block_builder):  
    block = block_builder(size)  
    blocks.append(block)
```

```
add_block(blocks, size=11,  
          build_triangle)
```

if ... elif ... else chain is often the smell of independent plug-in blocks

```
def add_block(blocks, size, type):  
    if type == 'circle':  
        # build circle of given size  
        block =   
    elif type == 'square':  
        # build square of given size  
        block =   
    elif type == 'triangle':  
        # build triangle of given size  
        block =   
    blocks.append(block)
```

```
add_block(blocks, size=11,  
          type='square')
```

```
def build_circle(size):  
    """ Build a circle. """  
    return 
```

```
def build_square(size):  
    """ Build a square. """  
    return 
```

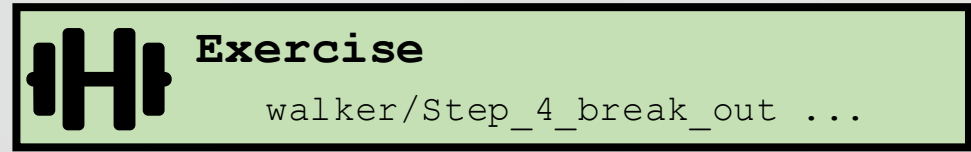
```
def build_triangle(size, angle):  
    """ Build a triangle. """  
    return 
```

```
def add_block(blocks, size, block_builder, kw_args):  
    block = block_builder(**kw_args)  
    blocks.append(block)
```

```
add_block(blocks, size=11,  
          build_triangle, {'angle': 35})
```

Conceptual Exercise

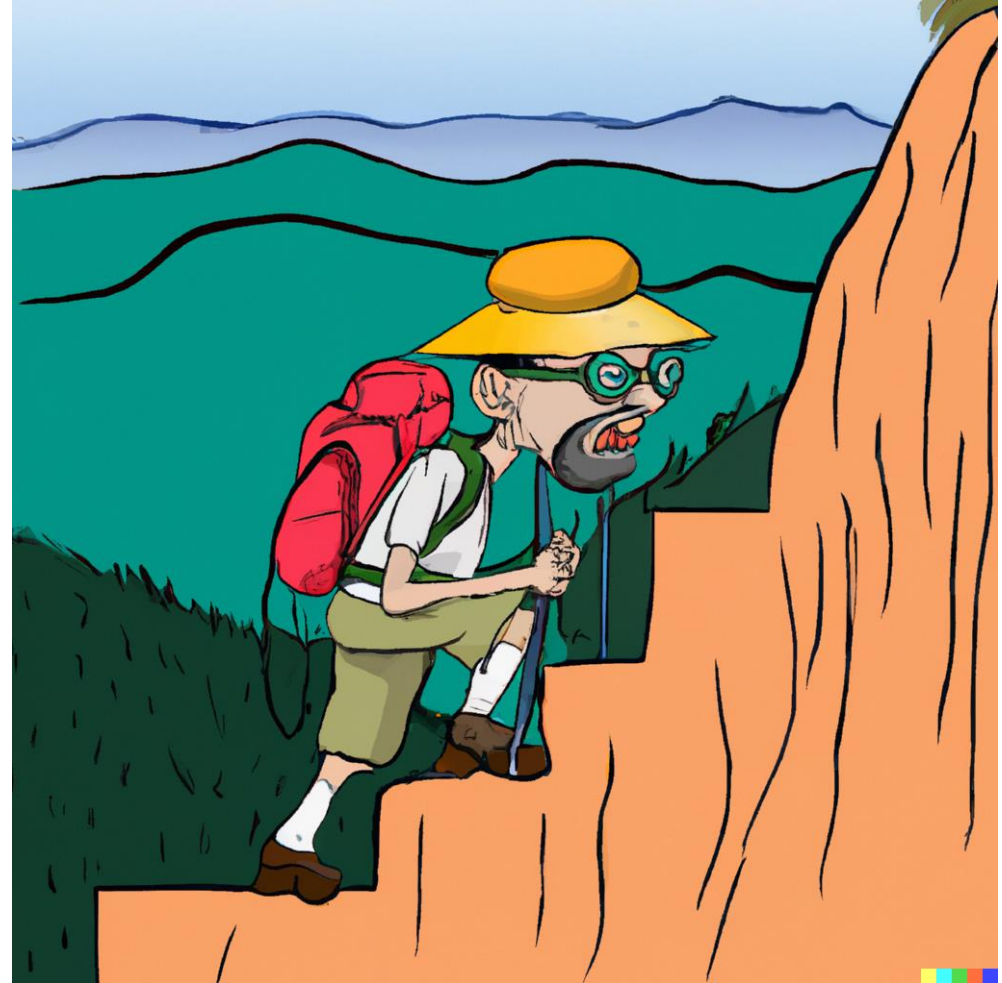
Two next-step proposals



- Go to the Notebook and follow the instructions there
- Comment on Issue #4 on git.aspp.school, suggesting what you think the code should look like.

The Walker: what have we achieved?

- We can run simulations with **different combinations** of context maps and next step proposals
- New context maps and next step proposals can be **contributed** by external people **without changing the code** in your package, or even knowing how it works
- We achieved **flexibility** and **openness to change**



Chapter 3: Separate what varies at the level of projects

The holy trinity of scientific computing

1. Provenance
2. Reproducibility
3. Organization



The holy trinity of scientific computing

1. Provenance

It needs to be clear **where**
data and plots come from,
when and **how** they were
generated

2. Reproducibility

All information necessary to
get the same result needs to
be saved

3. Organization

For your own sanity, you
should have a **consistent**
system for all this **data** and
artefacts



1. Provenance

- Data Provenance, or lineage, documents **where** data comes from
- Recording **when** and **by which** code the dataset has been changed
- But HOW?
 - External software? Usually not specific for scientific use case.
 - Folder structure/Filenames?
 - Code generated meta-information files? (see next slide)
- In case of plots:
 - data that generated the plots
 - For work in progress plots: `plt.annotate()`
 - Save .ipynb as pdf (with all the paths/version information in the notebook)
 - Could use metadata in images <https://github.com/dfm/savefig>

2. Reproducibility

- Save all information necessary to get the same result again
 - All input parameters to the code
 - Randomness- if used, save the seed
 - Which version of the code was used?
- Serialization of intermediate steps in the code (estimation and analysis)

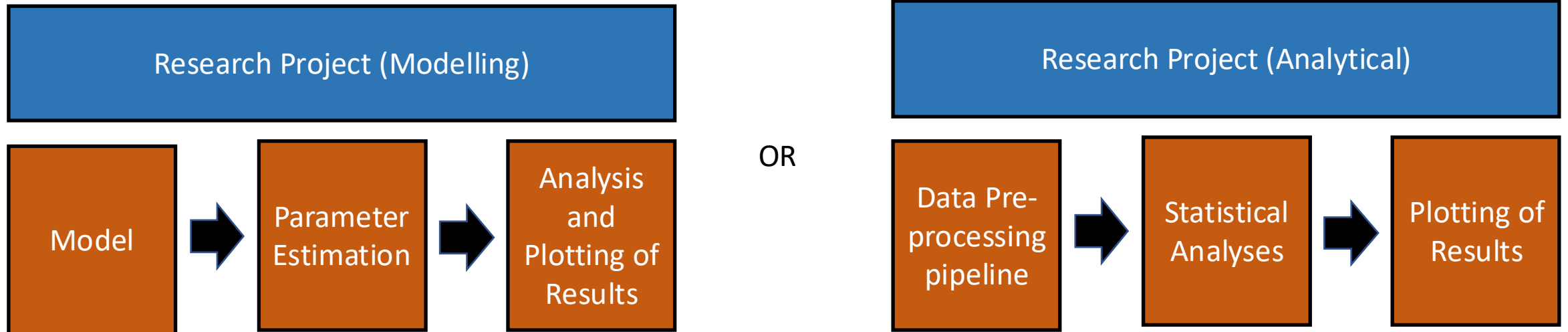
```
1  import git
2  import time
3
4  curr_time = time.strftime("%Y%m%d-%H%M%S")
5
6  repo = git.Repo(search_parent_directories=True)
7  sha = repo.head.object.hexsha
8
9  with open('meta.txt', 'w') as f:
10     f.write(f'I estimated parameters at {curr_time}.\n')
11     f.write(f'The git repo was at commit {sha}')
```

2. Reproducibility

- We also recommend using file names containing a version number or a time stamp, so that two subsequent runs do not overwrite previous results

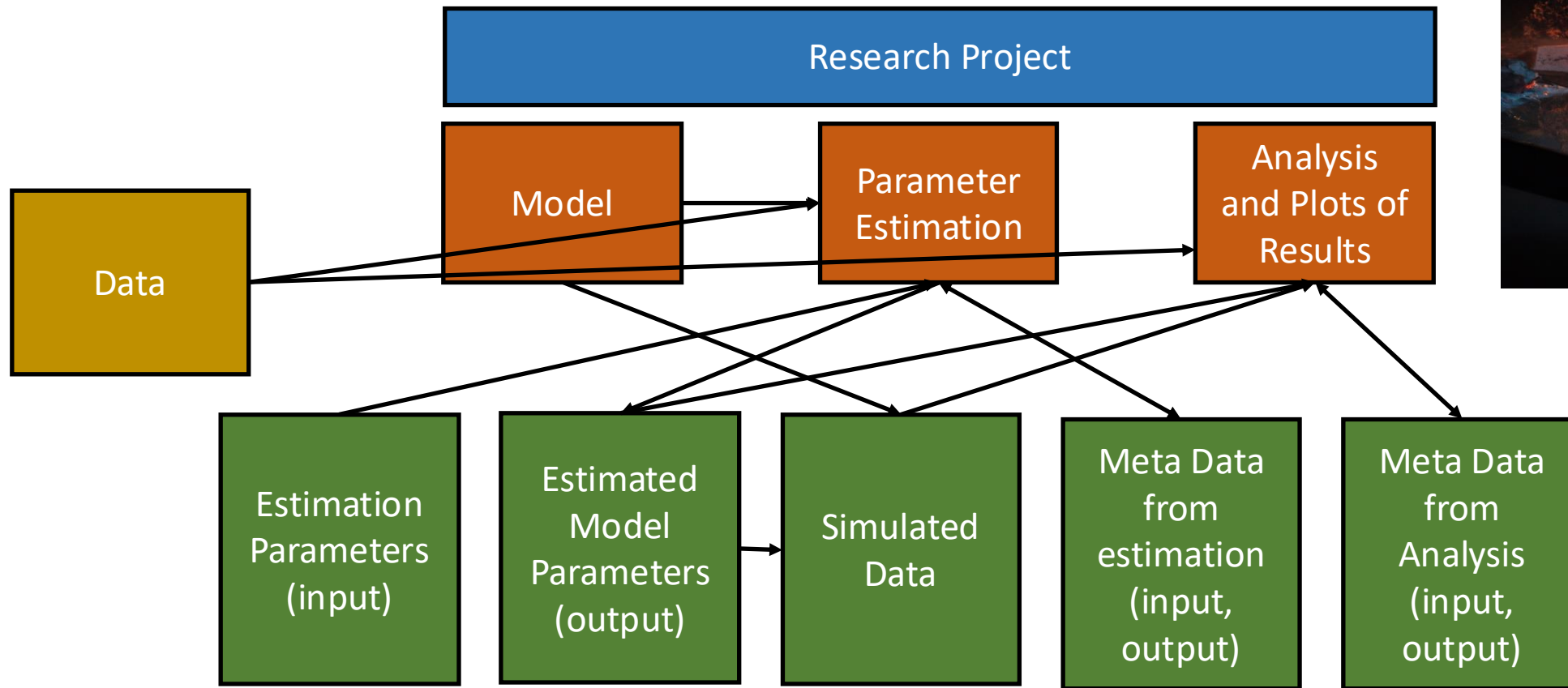
```
34     # STEP 4: Save the trajectory
35     curr_time = time.strftime("%Y%m%d-%H%M%S")
36     np.save(f"sim_{curr_time}", trajectory)
```

3. Organization



- Your research project may look something like this (or maybe you have different steps or a subset of steps)
- In any case it is likely that you will go through the steps many times before you're ready to publish your work

3. Organization

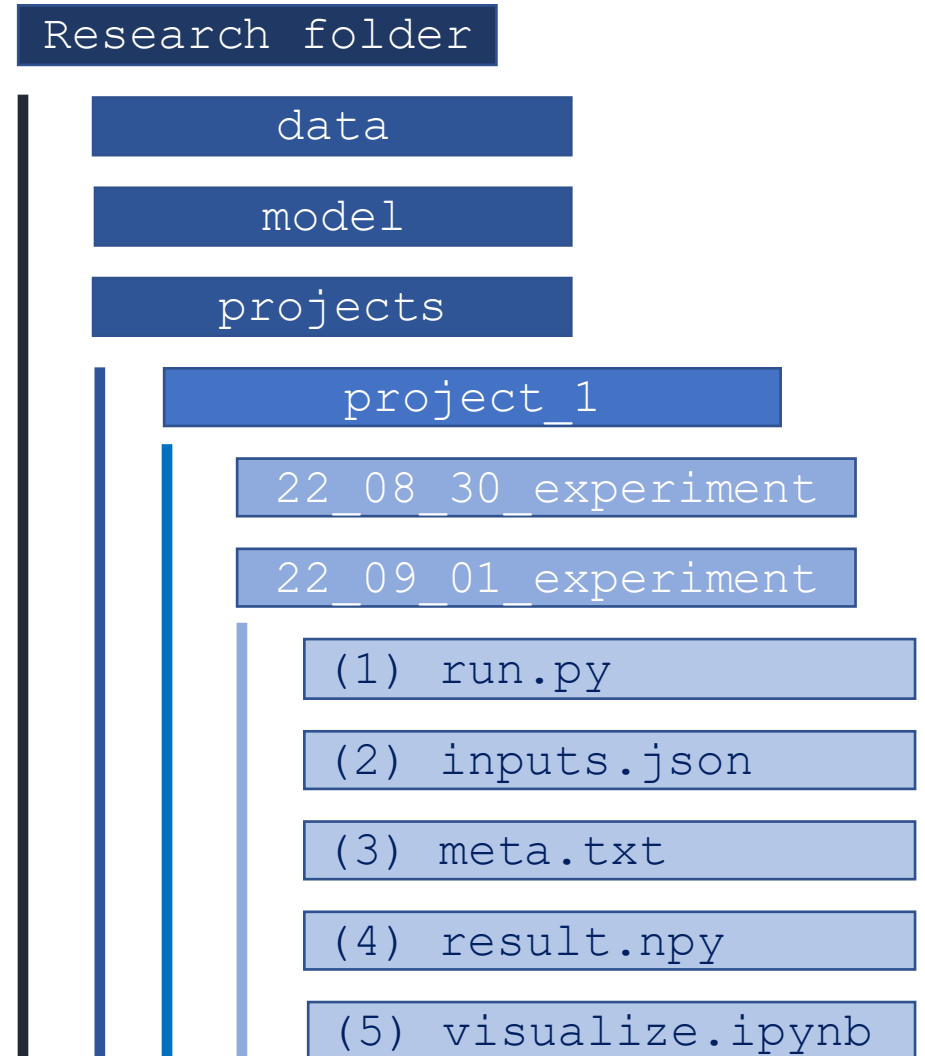


Each run has a bunch of associated data resulting folders and folders of data where no one knows which version of the code generated it or is using it!

3. Organization

Suggestion:

- Data should always be separated from code
- The model or algorithms or things that are applied to your data should be packages (see packaging lecture)
- Think of “runs” as Experiments. Each experiment has its own folder. The folder contains:
 1. **Minimal code** that calls the model and saves the result
 2. All **inputs** necessary to produce the result saved separately
 3. **Meta information** about which version of your code was used (and maybe a note about what you were trying to achieve, if you want to be extra nice to future you)
 4. The **result**
 5. Maybe the visualization of the result

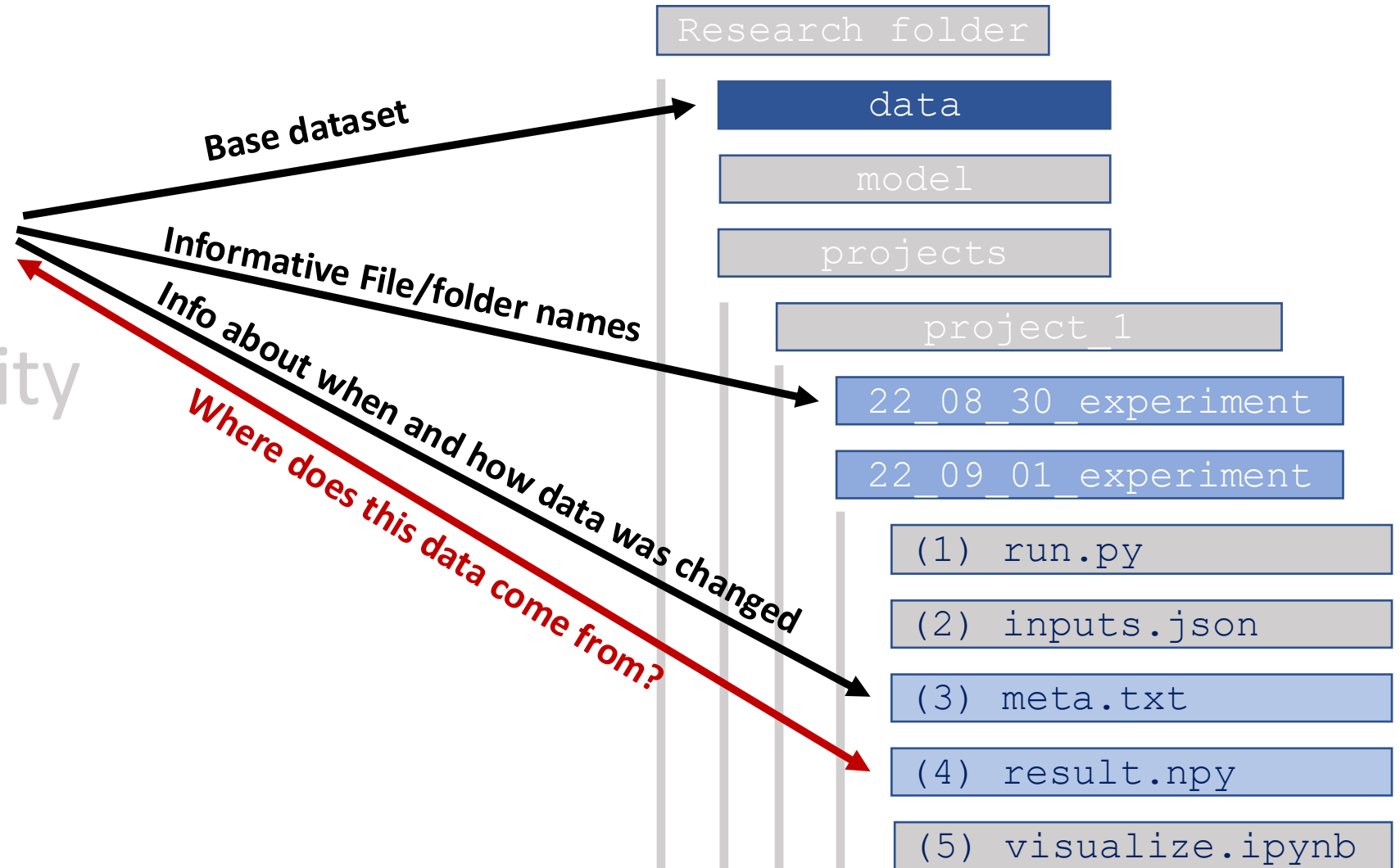


Organization

1. Provenance

2. Reproducibility

3. Organization

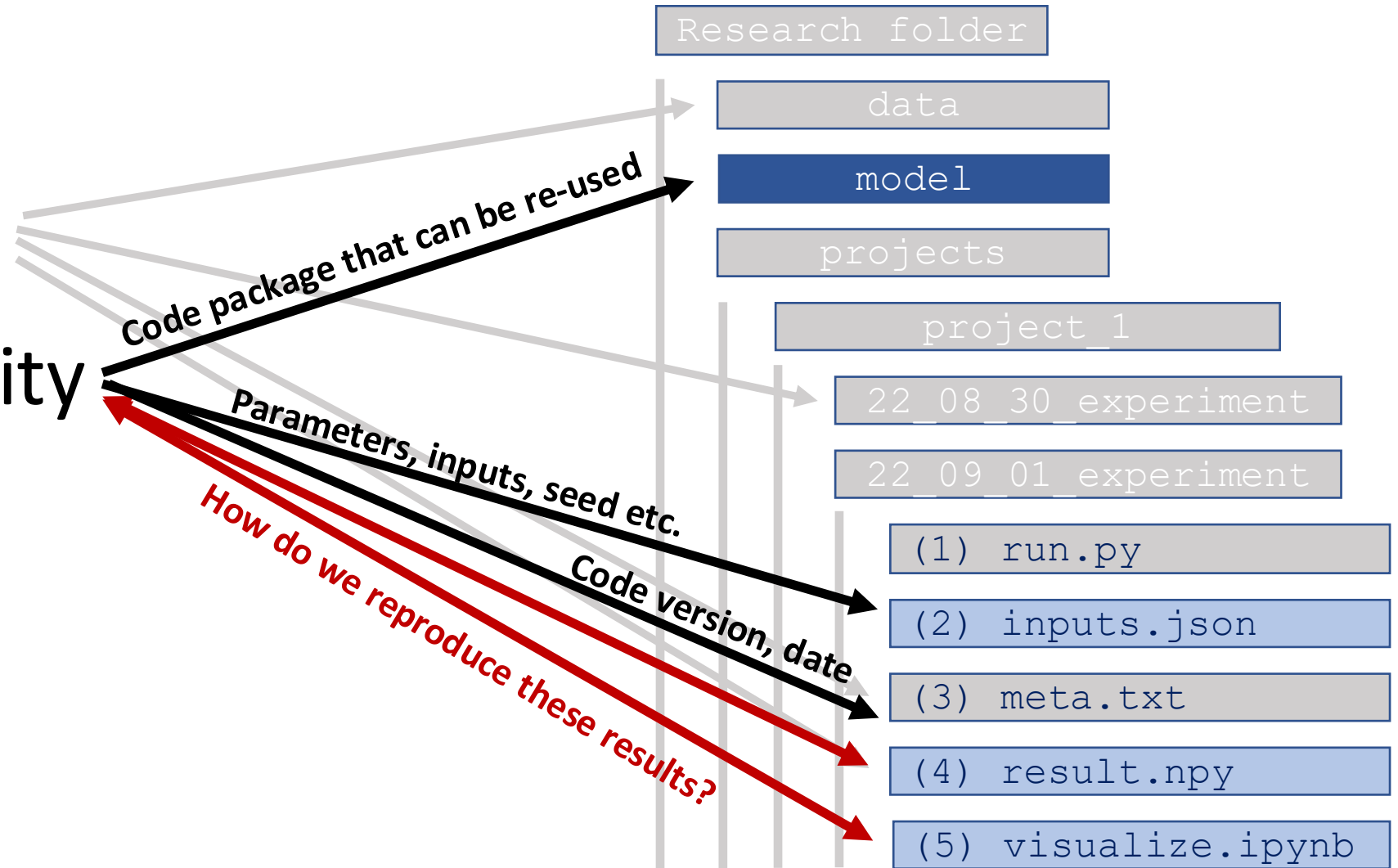


Organization

1. Provenance

2. Reproducibility

3. Organization

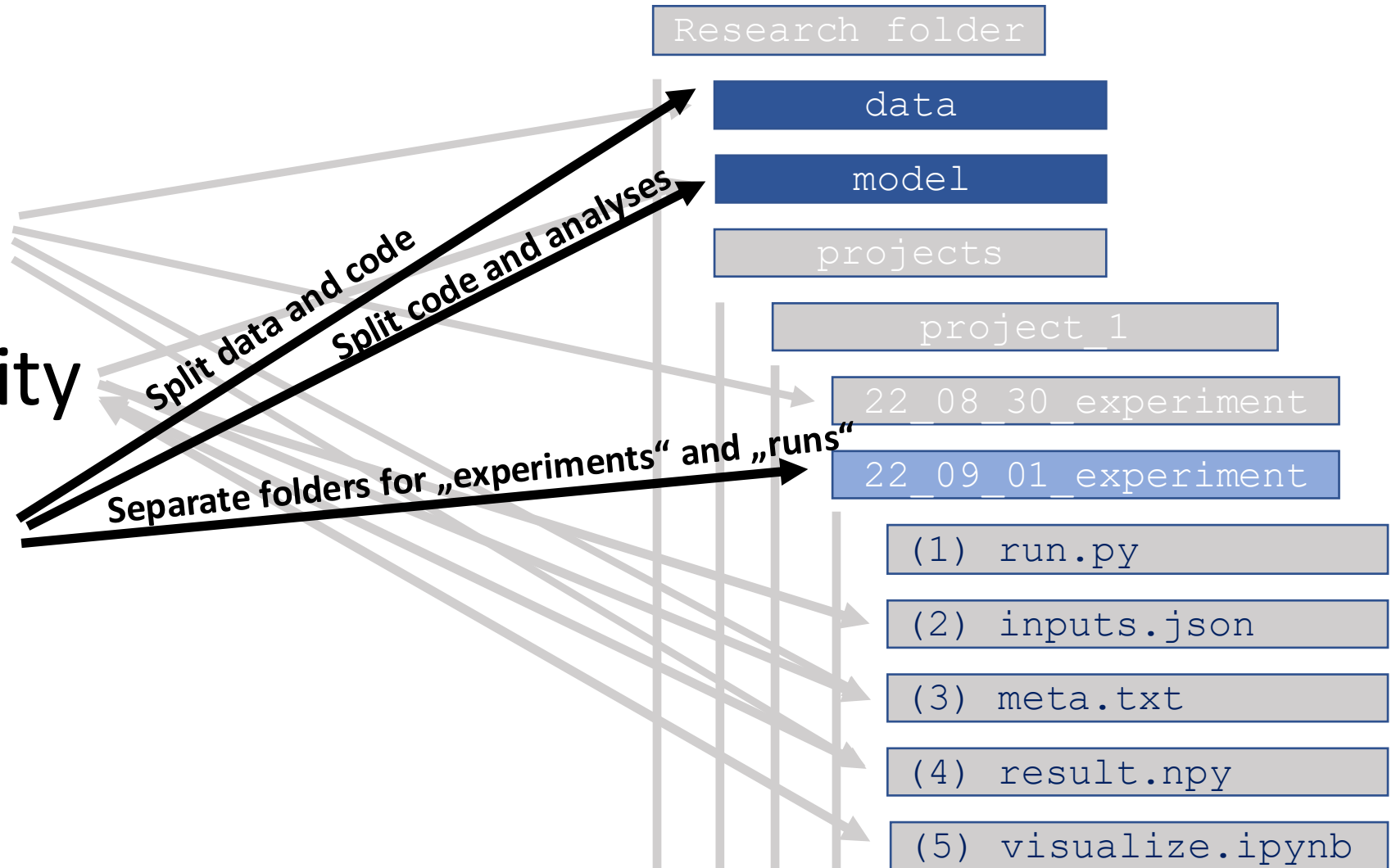


Organization

1. Provenance

2. Reproducibility

3. Organization



Exercise

Provenance and Reproducibility



Exercise

walker/Step_5_reproducibility/

Go through the following steps:

1. Complete the run.py script
2. Save the trajectory using ``np.save()``, and also save some metadata
3. Run the run.py script twice and confirm the results are identical by plotting them using the provided notebook
4. Submit PR for Issue #5 on GitHub



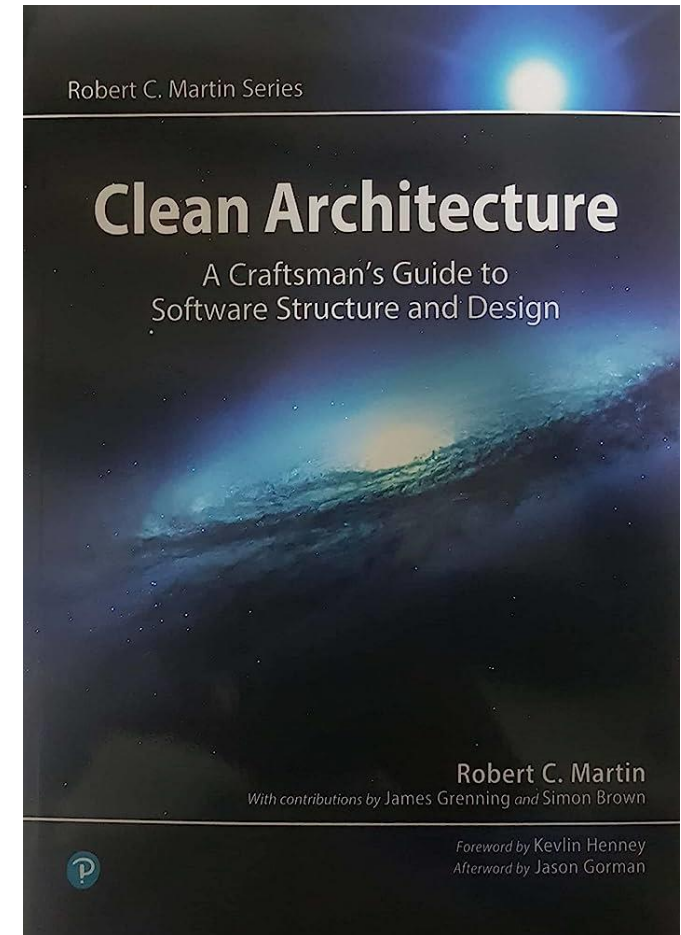
Live Coding

walker/Step_6_load_parameters_...

Provenance and Reproducibility

Where to go from here...

- Trust your nose! When your code smells, spend some time figuring out where the smell come from
- Don't get carried away: over-engineering counts as premature optimization



Thank you!



